Natalie Leaist

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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Engineering Physics*, minor in Honours Math* *Sept. 2023 - May 2029*

- 94.5% cumulative average (4.33/4.33 GPA)
- Trek Excellence Scholarship recipient (top 5% of faculty and year) & Dean's Scholar

Technical Skills

Programming Languages: Python, C, C++, Java, Javascript
Embedded Systems: STM32, TMS570, FreeRTOS, I2C, SPI, ethernet (UDP, lwIP), ADCs, sensors, Arduino
Data Analysis & Modelling: NumPy, matplotlib, Jupyter, FFTs, basic statistical tests, MATLAB, Simulink
Electrical: KiCad, PCB design, soldering, oscilloscope, function generator, multimeter
Other Tools: Git, Github, GitLab, VSCode, IntelliJ, Linux/WSL, OpenCV, LaTeX

Experience

Research Intern May 2025 – August 2025
Experimental Cosmology Group, UBC Physics & Astronomy *Vancouver, BC*

- Worked on the CGEM project, a radio telescope that will be used to help detect gravitational radiation from the early universe (collaboration between UBC, NASA Goddard Space Flight Centre and DRAO)
- Developed a high-frequency (830 Hz) embedded data acquisition system for accelerometer-based vibration and inclination monitoring of the telescope using an STM32 microcontroller, AD7177-2 ADC, DMA, SPI communication, FreeRTOS, and Ethernet (UDP/LwIP) streaming
- Performed Python-based data analysis, including FFTs, to verify accelerometer performance and calibrate sensor offset/gain
- Created a calibration procedure for accelerometer offset and gain
- Produced detailed documentation in LaTeX

Firmware Developer and Control Systems Team Member Sept. 2024 – Present
UBC Orbit Satellite Design Team *Vancouver, BC*

- Working on the ALEASAT project, an Earth observation satellite supported by the European Space Agency's Fly Your Satellite! Program
- Working on a team of five to develop a Helmholtz cage to simulate the magnetic field of low Earth orbit
- Modelling control systems using MATLAB and Simulink
- Developed firmware and software to fulfill mission and testing requirements set by the European Space Agency
- Implemented CPU usage monitoring across FreeRTOS tasks

Outdoor Adventure Guide July 2022 – August 2023
The Adventure Group *Whistler, BC*

- Led groups of up to 20 guests on Canada's highest and 2nd longest zip-line
- Conducted safety inspections on zip-lines, aerial obstacle courses, and off-road vehicles
- Trained in leadership, emergency response, risk assessment, and guest management

Projects

Self Balancing Robot

- Designed and built a two-wheeled self-balancing robot using Arduino, MPU6050 IMU, DRV8825 motor drivers, and NEMA-17 stepper motors powered by a LiPo battery
- Implemented a complementary filter to combine accelerometer and gyroscope data, providing a stable pitch estimate
- Developed and tuned a PID controller to maintain balance

Wordle Solver

- Developed a Python algorithm capable of solving Wordle puzzles with an average of approximately 4 guesses
- Implemented a word-elimination strategy to reduce the number of possible solutions after each guess
- Used object-oriented programming in Python to organize the solver into modular, reusable classes

Activities

Freeride Skiing

- Ranked 2nd in Canada in my age category (2022-2023 season)
- Qualified for Junior World Championships (2023) and competed at the North American level
- Sponsored by Rossignol Freeride (Sept 2023 – Aug 2024)

Certifications

- UBC PHAS Student Machine Shop Course (35 Hours)
- Class 5 Drivers' License
- 16 Hour Standard First Aid
- PMBIA Level 1 Mountain Bike Instructor